ENG3042
Propulsion and Turbomachinery 3
Lab Report

Fatimah Hasan
2888172h
School of Engineering
University of Glasgow

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0.1 Introduction

This laboratory experiment aims to measure the noise of a small jet engine (Cussons) at 4 stages: idle, 30% , 50% ,and 70% throttle. The Sound–Pressure–Level(SPL) of the jet engine was detected by a calibrated high–precision acoustic Microphone (Rion NL-53) and analysed using Fourier analysis. The SPL was found by analysing the frequency domain, which illustrates the link between sound and jet velocity using MATLAB software for the full analysis.

0.2 Method

The experiment is not done in an acoustics lab, so the assessment is qualitative. The used Microphone converts sound waves (dB) to a voltage waveform using MATLAB.

0.2.1 Verification of method:

Following the Nyquist sampling theorem, which states that the sampling frequency ($\mathbf{f_s}$) must be at least twice the signal frequency (f) to avoid data corruption ($\mathbf{f_s \geq 2f}$). The connected signal frequency (f) of the microphone generates a pure signal of f = 20000Hz and fs = 40000Hz, the values acquired using the **calibration.dat** file, which is a table with 2 rows. Each data is in a (row x column) 2xN array, where N is 40000 sample. However, applying the Fourier analysis to the signal in MATLAB confirms the Sine wave in the Time Domain forming 10 cycles and the pure 1Hz spick on the frequency domain, as in Figure 1.

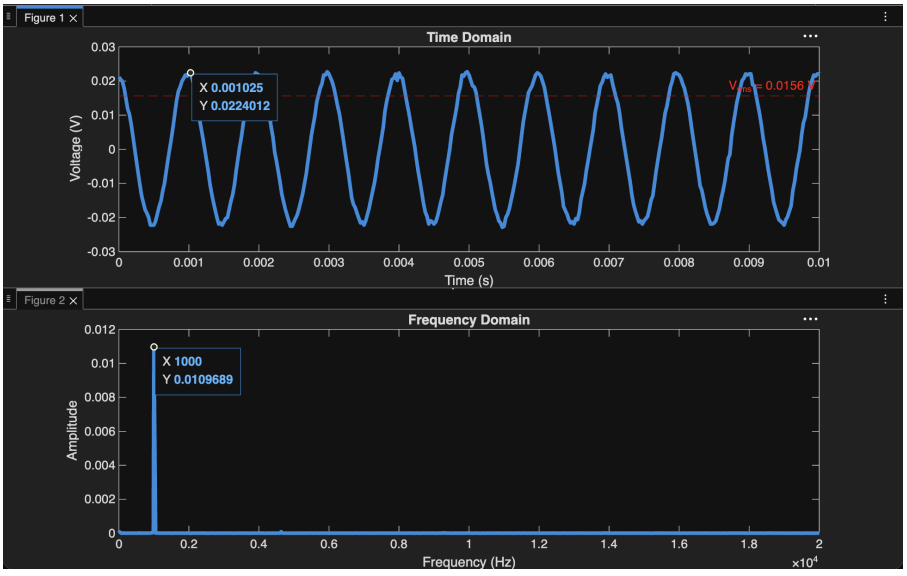


Figure 1: Top figure is Time domain Sine wive and lower figure shows the Frequency Domain main hormanic

0.3 Results and Discussion

Now evaluating the microphone Sensitivity S from the calibration data from the following:

1. Finding the Pressure (P) level as decibel, dB. which given by

$$\mathrm{SPL} = 20\log_{10}\left(\frac{p_{\mathrm{rms}}}{p_{\mathrm{ref}}}\right), \qquad p_{\mathrm{ref}} = 20\,\mu\mathrm{Pa} \tag{1}$$

- $SPL = 94$ dB decimal of pure sine wave of 1Hz.
- Rearrange equation (1),

$$p_{\text{cal}} = p_{\text{ref}} \times 10^{\text{SPL}/20} = 1.0024(\text{Pa}) \tag{2}$$

2. Root Mean Square (RMS) voltage V_{rms} ,

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{0.0156}{\sqrt{2}} = 0.0156\ V/Pa \tag{3}$$

3. Calibration of the Microphone sensitivity S,

$$S = \frac{V_{\text{rms}}}{p_{\text{rms}}} = \frac{0.0156}{1.0024} = 0.0155 V/Pa \tag{4}$$

4. Converting the voltage of Turbojet test data of Ideal 0%,30% ,50% ,70%, to pressure using sensitivity S by rearranging equation 4 and following the same method as in steps 2 and 1, which gives the results in Table 1

Table 1: Measured RMS voltage, RMS pressure and Sound Pressure Level SPL

File	V_{rms} (V)	p_{rms} (Pa)	SPL (dB)
`test0.dat`	0.1249	8.018	112.1
`test30.dat`	0.1225	7.862	111.9
`test50.dat`	0.2400	15.408	117.7
`test70.dat`	0.3254	20.889	120.4

5. Figure 2 shows the attempt to apply Fourier analysis, which shows the frequency spectra for the four data test sets. Each one has one dominant tone, the compressor blade passing frequency that increases the speed of the engine considering the blades' angle of attack [1]. The results show at Ideal frequency is 8523 Hz (3.2 Pa), 10659 Hz (4.3 Pa) at 50% and 11551 Hz (7.1 Pa) at 70%. The Idea is mostly pure tone.

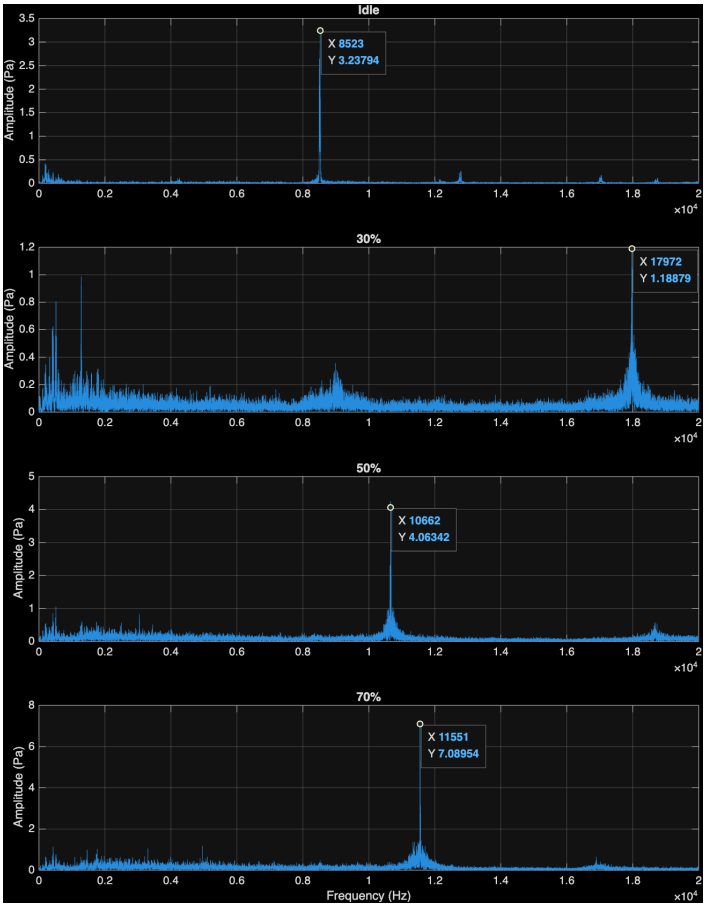


Figure 2: Fourier analysis analysis of the turbojet noise signal.

0.4 Conclusion:

The calibrated microphone was with a 94 dB,1kHz tone, with sensitivity of S=15.5 mV/Pa and $V_{\text{cal}} = 15.6mV/Pa$. The Fourier analysis verified this by recovering the 1KHz tone from the calibrated data tuple. The overall SPL increases from 112.1dB at ideal to 120.4dB at 70% engine thrust. The compressor blade passing tone was dominating the Idle values test. At the higher speed, the jet noise level grew steadily, knowing that the collected data is not 100% acurat due to the room reflection of the noise, distance measurement and veriatry of noise sources.

Bibliography

[1] M. J. T. Smith, *Aircraft Noise*. Cambridge: Cambridge University Press, 1989.